

# The Time Travel Diary

**II PUC · Ch 6 · Evolution**

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer





# Back to the Beginning

Prof. BAIO's time machine roared back billions of years. "The universe began with the Big Bang. Earth formed about 4.5 billion years ago. **Oparin** and Haldane said life arose from non-living molecules — **chemical evolution**. In 1953, **S.L. Miller** made amino acids in a flask using sparks!"

> 💡 **REMEMBER**
> **Miller used $CH_4$, $H_2$, $NH_3$ and water vapour at 800 °C with electric discharge.**



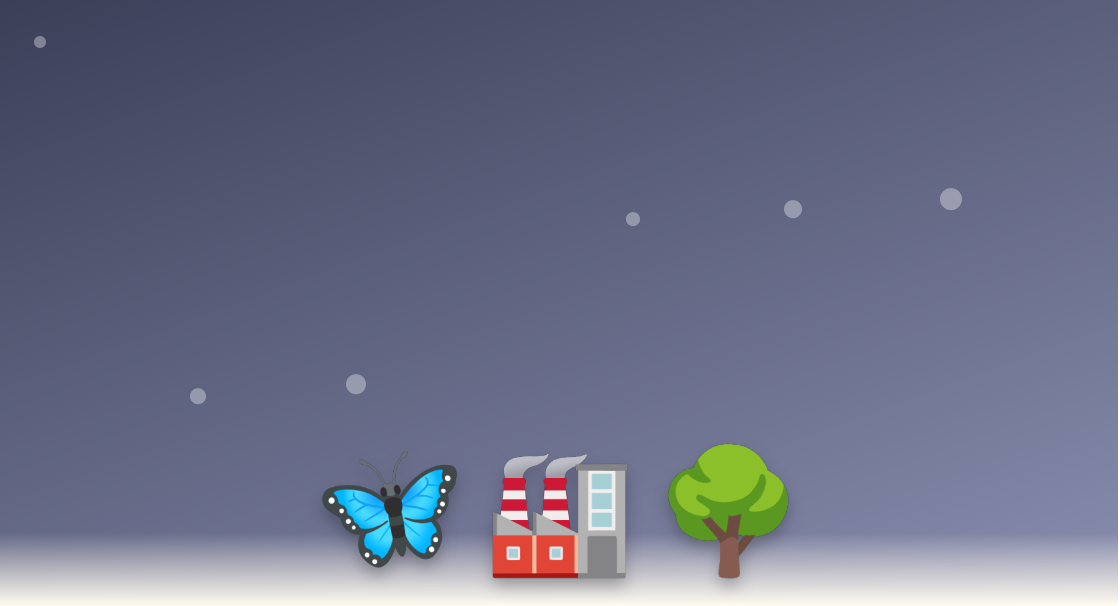
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# Clues in Bodies

Back in the lab, Bindu compared limbs. The forelimbs of a whale, a bat and a human have the same bones — **homologous organs**, from **divergent evolution**. The wings of a butterfly and a bird do the same job but are built differently — **analogous organs**, from **convergent evolution**.

💡 REMEMBER

**Thorn of Bougainvillea and tendril of Cucurbita are homologous; sweet potato and potato are analogous.**



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# The Moth Mystery

In England after 1850, factory soot darkened the tree trunks. White moths were easy for birds to spot, but dark moths hid well. Soon dark moths were the majority — **industrial melanism**, natural selection in action!

💡 REMEMBER

**Antibiotic resistance and herbicide resistance are also evolution caused by human activity.**





# Darwin’s Voyage

They sailed with Charles Darwin on the HMS Beagle to the Galápagos Islands. Darwin's finches had different beaks for different foods — all from one ancestor. That is **adaptive radiation**. Australian marsupials did the same.

💡 **REMEMBER**

**Placental mammals and Australian marsupials look alike — convergent evolution.**





# Survival of the Fittest

"Living things vary, and they produce more young than can survive," Darwin wrote. "Those best suited to the environment leave more offspring." This is **natural selection**. **Fitness** really means reproductive fitness. Alfred Wallace reached the same idea.

💡 REMEMBER

**Darwin's key ideas: branching descent and natural selection.**

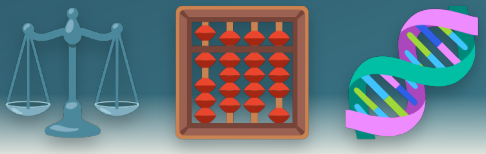




# Sudden Jumps

Hugo de Vries studied the evening primrose. He said new species come from large, sudden changes — **mutations** — not small steps. He called this single-step jump **saltation**.

> 💡 **REMEMBER**
> For de Vries, mutations are random and directionless.



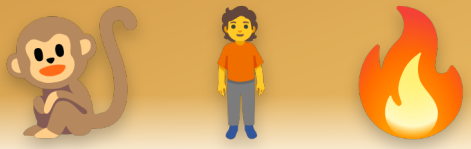


# Keeping the Balance

The **Hardy–Weinberg principle** says allele frequencies stay constant: $p^2 + 2pq + q^2 = 1$. Gene flow, **genetic drift**, mutation, recombination and natural selection disturb it. When a few individuals start a new population, drift causes the **founder effect**.

💡 REMEMBER

**Natural selection can be stabilising, directional or disruptive.**





# Our Family Tree

Last stop — our own story! Dryopithecus and Ramapithecus lived about 15 million years ago. **Homo habilis** (650–800 cc brain) probably did not eat meat. **Homo erectus** (900 cc) probably did. **Neanderthal man** (1400 cc) buried the dead. Finally, **Homo sapiens** arose in Africa and spread across the world.

💡 **REMEMBER**

**Agriculture began about 10,000 years ago and human settlements started.**



KEY TERMS

# Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Oparin p.1 · chemical evolution p.1 · S.L. Miller p.1 · homologous organs p.2 · divergent evolution p.2 · analogous organs p.2 · convergent evolution p.2 · industrial melanism p.3 · adaptive radiation p.4 · natural selection p.5 · Fitness p.5 · mutations p.6 · saltation p.6 · Hardy–Weinberg principle p.7 · genetic drift p.7 · founder effect p.7 · Homo habilis p.8 · Homo erectus p.8 · Neanderthal man p.8 · Homo sapiens p.8



QUICK QUIZ

# Do you remember?

**1. Miller's experiment produced:**

- Proteins
- Amino acids
- DNA
- Cells

**2. Forelimbs of whale, bat and human are:**

- Analogous
- Homologous
- Vestigial
- Fossils

**3. Darwin's finches show:**

- Industrial melanism
- Adaptive radiation
- Saltation
- Genetic drift

**4. Saltation was proposed by:**

- Darwin
- Lamarck
- Hugo de Vries
- Wallace

**5. Brain capacity of Homo erectus was about:**

- 650 cc
- 900 cc
- 1400 cc
- 1600 cc

Answers: 1-Amino acids · 2-Homologous · 3-Adaptive radiation · 4-Hugo de Vries · 5-900 cc